

PAIRSELECT: TRAINING-FREE TOKEN SELECTION FOR EFFICIENT WEAKLY SUPERVISED VIDEO ANOMALY DETECTION

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ABSTRACT

Detecting rare events requires sustained observation, but not necessarily full computation for every internal video token. We introduce PairSelect, a training-free token selection operator for frozen weakly supervised video anomaly detectors. After dense contextualization in the first half of a video encoder, PairSelect allocates local quotas and retains activation-magnitude-guided representatives without averaging their vectors. It uses adjacent spatial pairs for joint-attention encoders and complete spatial trajectories for divided space–time attention. Neither input clip coverage nor the dense-trained detection head is changed. Across three encoders and two datasets, the completed middle-budget evaluation removes approximately 40–43% of subsequent tokens, with primary-metric changes from -0.16 to $+0.68$ percentage points. Analytical accounting predicts 21.75–23.04% fewer operations in the twelve principal transformer blocks. Additional budgets and measurements of full-encoder FLOPs, latency, throughput, and peak GPU memory remain to be completed.

Index Terms— video anomaly detection, weak supervision, token selection, training-free compression, efficient inference
Author working draft: unmeasured results are marked with dashes; author revision and verification are required before submission.

1. INTRODUCTION

Video anomaly detection faces an asymmetric computational problem: unusual events are scarce, yet a system must process long stretches of ordinary activity to find them. In weakly supervised video anomaly detection (WSVAD), video-level training labels do not identify the anomalous snippets, which can be overwhelmed by normal instances [1, 2]. A practical pipeline extracts clip representations with a pretrained encoder and passes the resulting sequence to a temporal detector [3]. Reducing the number of input clips is one way to lower cost, but it also changes the temporal observations available to that detector. We instead ask how to reduce internal computation while keeping the original clip schedule intact.

The rarity of anomalous events does not imply that every internal token carries unique information. VideoMAE demonstrates the usefulness of video redundancy for masked pretraining [4]; token sampling and merging also exploit redundant representations in pretrained transformers [5, 6]. These findings motivate a different allocation of inference effort: allow every token to participate in the first half of the encoder, then continue processing only a structured subset. Pretraining redundancy alone does not establish that a frozen anomaly detector tolerates this intervention. The resulting representations must still work with its existing detection head.

We introduce PairSelect, a local, non-mixing selection operator for this setting (Fig. 1). It combines three choices: spatially adjacent candidate units, fixed group budgets, and an activation-magnitude preference within those units. For joint-attention encoders, a first selection round gives each pair a representative whenever the budget permits; remaining slots retain additional members of stronger pairs. For divided space–time attention, entire spatial trajectories are selected together. The selected vectors are gathered without averaging at the insertion point. No selector is trained, no compression-specific labels are required, and neither the encoder nor the dense-trained head is fine-tuned after insertion.

Our contributions are a training-free, structure-aware selection rule for frozen WSVAD pipelines; explicit budget and computational-cost accounting for joint and divided attention; and a direct-insertion evaluation on three encoders and two datasets. At the completed middle budget, the largest observed primary-metric decrease is 0.16 percentage points, while approximately 40–43% of tokens are removed from the remaining encoder blocks. The budget ladder separates local duplicate removal from more aggressive loss of local representatives, providing a concrete basis for studying quality–cost trade-offs rather than assuming a preferred compression ratio.

2. RELATED WORK

Weakly supervised anomaly detection. Multiple-instance ranking [1], robust temporal feature magnitudes [2], and uncertainty-regulated dual memories [3] address video-level supervision. VadCLIP adapts frozen vision–language representations through a task-specific detection architecture [7]. Efficient anomaly detectors also include learned autoencoder designs [8]. We use UR-DMU as a fixed downstream head within each encoder–dataset configuration, and reduce visual encoding cost rather than introduce another detector. Our setup is visual-only even on the audiovisual XD-Violence benchmark [9].

Efficient transformers. VideoMAE and VideoMAEv2 use masked video pretraining [4, 10]; TimeSformer separates temporal and spatial attention [11]. DynamicViT learns token-importance predictors [12], and EViT reorganizes attentive and inattentive tokens [13]. EVAD studies task-aware token dropout for action detection [14]. ATS provides parameter-free sampling, and ToMe supports merging without additional training [5, 6]. Thus, training-free reduction itself is not new. PairSelect studies a local non-mixing rule at a fixed intermediate depth with an unchanged anomaly head and unchanged clip coverage. ESOM and LAVIDA instead compress tokens in multimodal language-model pipelines [15, 16], unlike our frozen visual-encoder/MIL setting.

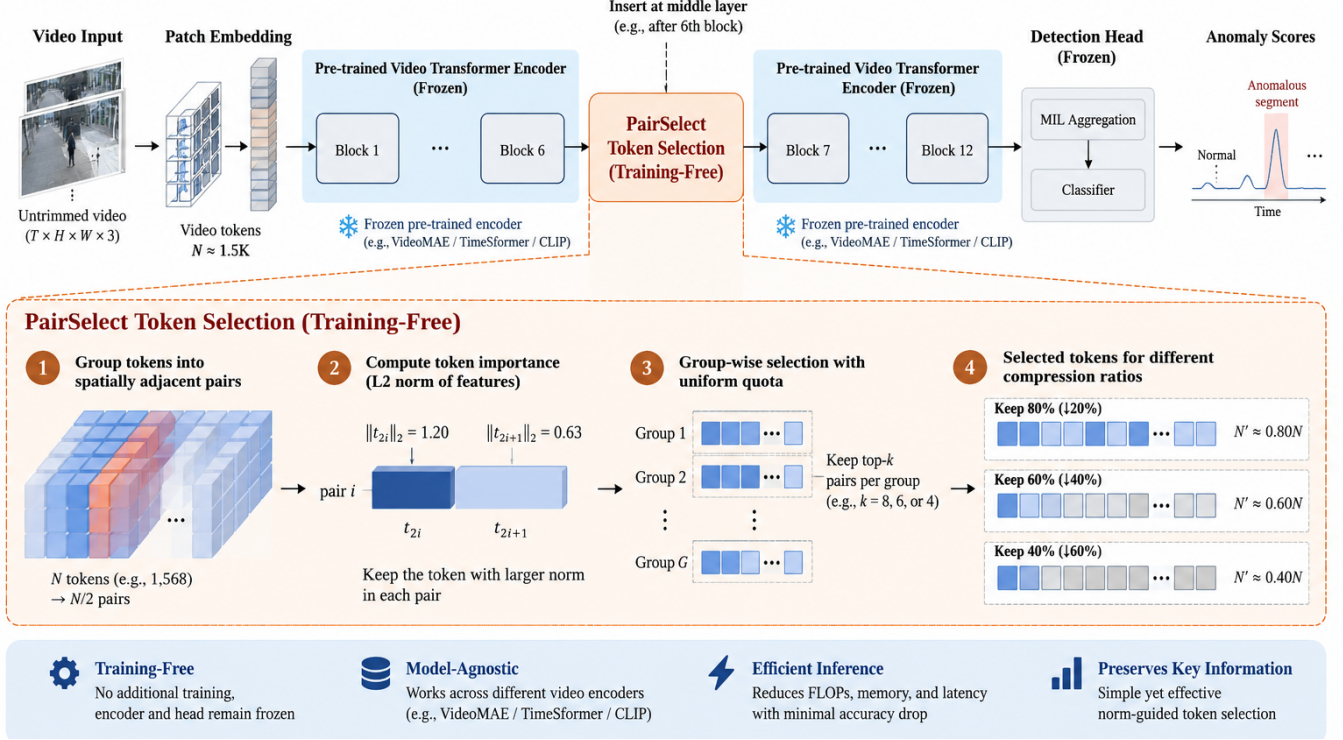


Fig. 1. Overview of PairSelect. Six dense blocks contextualize all tokens before local, magnitude-guided selection reduces the input to blocks 7–12. Group quotas control pair-member selection; divided attention retains complete spatial trajectories. Clip coverage and the dense-trained detection head remain unchanged.

3. PAIRSELECT

3.1. Frozen pipeline and design rationale

Let $E_{1:b}$ and $E_{b+1:L}$ denote the prefix and suffix of a pretrained encoder, R its clip readout, and H the temporal detection head. For scheduled clips $\{v_t\}_{t=1}^M$, inference is

$$z_t^{(r)} = R(E_{b+1:L}(S_r(E_{1:b}(v_t)))), \quad \hat{y}^{(r)} = H(\{z_t^{(r)}\}_{t=1}^M). \quad (1)$$

Only S_r , the selection operator at retention budget r , is inserted. We set $L = 12$ and $b = 6$ for all three encoders. The clip count M , output dimension, and head weights remain unchanged. Training-free refers to compression: the original detection head is trained once on dense features for each configuration.

Local redundancy motivates nearby comparisons; gathering existing vectors avoids additional mixing before the unchanged

head. ATS motivates inexpensive magnitude-based scoring [5], but our block-output L2 norm differs from its attention-weighted value norm. High norm is not an anomaly probability and may occur in background regions [17]; local quotas therefore constrain magnitude preference rather than treating it as a global anomaly oracle.

3.2. Local quotas and pair-member priority

For VideoMAE and VideoMAEv2, the block output is $X \in \mathbb{R}^{N \times D}$ with a known patch grid. We form nonoverlapping horizontal pairs, ordered by native coordinates. Consecutive groups contain at most ten tokens (five pairs). A group with n_g tokens retains

$$q_g(r) = \lfloor r n_g + \frac{1}{2} \rfloor, \quad r \in \{0.8, 0.6, 0.4\}. \quad (2)$$

For pair j with members $(x_{j,0}, x_{j,1})$, define

$$a_j = \arg \max_{u \in \{0,1\}} \|x_{j,u}\|_2, \quad s_j = \max_{u \in \{0,1\}} \|x_{j,u}\|_2. \quad (3)$$

Ties retain native order. Let $w_j = x_{j,a_j}$ and $\ell_j = x_{j,1-a_j}$. If π sorts the m_g pairs by descending s_j , the group’s priority list is

$$\mathcal{P}_g = (w_{\pi_1}, \dots, w_{\pi_{m_g}}, \ell_{\pi_1}, \dots, \ell_{\pi_{m_g}}). \quad (4)$$

We take the first q_g entries, then restore the selected indices to native order before gathering. At $r = 0.8$ or 0.6 , every pair in a full group retains a representative; at $r = 0.4$, only four of its five representatives survive. This is a coverage property of the rule, not a guarantee about anomaly preservation. No averaging or rescaling is applied to a selected vector.

Table 1. Token budgets and analytical twelve-block FLOPs reduction. Deletion is nominal unless marked actual.

Delete	K	Actual delete	N/K	Block FLOPs ↓
<i>VideoMAE / VideoMAEv2</i>				
20%	1254	20.03%	$1.250 \times$	12.05%
40%	941	39.99%	$1.666 \times$	23.04%
60%	627	60.01%	$2.501 \times$	33.05%
<i>TimeSformer (including CLS)</i>				
20%	1233	21.41%	$1.273 \times$	10.94%
40%	897	42.83%	$1.749 \times$	21.75%
60%	561	64.24%	$2.797 \times$	32.43%

Table 2. PairSelect at nominal deletion budgets (metrics: 0–100). Δ_{40} and its paired 95% confidence interval (CI) are changes from dense in percentage points. Dashes denote unmeasured budgets; actual token counts are in Table 1.

Encoder	Dense	Delete 20%	Delete 40%	Delete 60%	Δ_{40}	95% CI of Δ_{40}
<i>UCF-Crime — frame ROC-AUC</i>						
VideoMAEv2-B	80.63	—	81.02	—	+0.39	[−0.94, +1.90]
VideoMAE-B	80.55	—	80.39	—	−0.16	[−0.75, +0.50]
TimeSformer-B	78.68	—	78.68	—	−0.001	[−1.12, +1.11]
<i>XD-Violence — frame trapezoidal PR-AUC</i>						
VideoMAEv2-B	76.18	—	76.51	—	+0.32	[−0.06, +0.75]
VideoMAE-B	70.34	—	70.94	—	+0.61	[+0.02, +1.17]
TimeSformer-B	76.69	—	77.37	—	+0.68	[+0.22, +1.13]

3.3. Divided attention and actual budgets

TimeSformer requires a different unit: all T tokens at a spatial location form a trajectory, scored by the maximum member norm. Each group of ten trajectories retains its top q_g trajectories. The current bridge requires the retained spatial count to be a multiple of the native patch width W ; we round the total downward and remove excess quota from tail groups. With $P = 196$, $T = 8$, and $W = 14$,

$$P' = W \left\lfloor \frac{\sum_g q_g(r)}{W} \right\rfloor, \quad K = TP' + 1, \quad (5)$$

where the CLS token is preserved. This width constraint belongs to the current implementation, not to all divided-attention models. Dynamic selections occupy a fixed quota-slot skeleton: gathered content is unchanged, but slot coordinates use approximate anchors rather than an exact reconstruction of the original lattice. Complete trajectories preserve their time indices; the geometry adjustment can empty tail groups.

3.4. Computational accounting

For joint attention with hidden width D and MLP ratio four, the principal block cost in multiply–accumulate operations (MACs) is

$$C_J(N, D) = 12ND^2 + 2N^2D. \quad (6)$$

The linear term includes QKV, output projection, and the MLP; only attention products scale quadratically in N . For standard divided attention, including the temporal projection and replicated CLS in spatial attention [11],

$$C_T(P, T, D) = (17PT + 4T + 8)D^2 + 2D\{PT^2 + T(P + 1)^2\}. \quad (7)$$

The twelve-block cost ratio is $\frac{1}{2}(1 + C_{\text{reduced}}/C_{\text{dense}})$. Table 1 uses $D = 768$ and counts one MAC as two FLOPs. It excludes embedding, readout, normalization, and selection costs. The selector adds norm computation, bounded-size sorting, and gathering; these may have non-negligible runtime despite low arithmetic cost. Token compression, full-encoder FLOPs, latency, and peak memory are therefore distinct quantities.

4. EXPERIMENTS

4.1. Setup and evaluation

We evaluate UCF-Crime [1] with 1,610 training and 290 test videos, and XD-Violence [9] with 3,950 accepted training videos and all

Table 3. Same-quota references at nominal 40% deletion: changes from dense in percentage points. Only PairSelect is proposed.

Encoder	PairSelect	Prefix	Random
<i>UCF-Crime</i>			
VideoMAEv2-B	+0.39	−0.11	+0.54
VideoMAE-B	−0.16	−0.43	−0.09
TimeSformer-B	−0.001	−0.33	+0.21
<i>XD-Violence</i>			
VideoMAEv2-B	+0.32	−0.41	−0.43
VideoMAE-B	+0.61	+1.14	+1.09
TimeSformer-B	+0.68	+0.10	−0.11

800 test videos. Four of XD’s 3,954 declared training videos are excluded after source-integrity checks. Only visual input and video-level training labels are used. Each of VideoMAEv2-B, VideoMAE-B, and TimeSformer-B is paired with a separately trained UR-DMU head [3]. Head training uses 200 temporal bins, 64 normal and 64 anomalous bags per step, Adam with learning rate 10^{-4} and weight decay 5×10^{-5} , and the final checkpoint after 3,000 steps. This is a controlled dense-feature training protocol, not a reproduction of the detector’s published I3D-feature scores.

The same head checkpoint, preprocessing, and clip schedule are reused across compression settings within a configuration. VideoMAE-family clips yield $8 \times 14 \times 14 = 1568$ tokens; TimeSformer yields the same patch count plus CLS. The completed runs preserve 69,364 UCF and 145,703 XD clips per VideoMAE-family encoder, and 138,853 and 291,781 clips for TimeSformer. Clip totals differ across encoders, but dense and reduced paths match exactly within each configuration. Model weights are frozen during compressed inference.

UCF uses frame ROC-AUC. XD uses trapezoidal PR-AUC under the fixed author-style 16-frame prefix alignment; non-interpolated average precision (AP) is reported separately. These are different integration conventions, not interchangeable names. Differences use 10,000 paired video-level bootstrap resamples stratified by video label. The intervals reflect test-video sampling uncertainty conditional on the reported runs, not variability across training seeds. The 20% and 60% deletion evaluations are a budget-sensitivity extension; no preferred operating point is inferred from missing results.

4.2. Direct insertion without head adaptation

Table 2 reports the completed middle-budget results. On UCF, PairSelect changes ROC-AUC by +0.39, −0.16, and approximately 0.00

Table 4. Efficiency of dense and PairSelect configurations. F_{blocks} is an analytical principal-block estimate (1 MAC = 2 FLOPs). Full-encoder FLOPs, latency, throughput, and peak allocated/reserved memory remain unmeasured.

Encoder	Delete	Tokens	F_{blocks} GFLOPs	F_{full} GFLOPs	Latency ms/clip	Throughput clips/s	Peak memory alloc./res. GiB
VideoMAEv2-B	Dense	1568	356.99	—	—	—	—/—
	20%	1254	313.99	—	—	—	—/—
	40%	941	274.74	—	—	—	—/—
	60%	627	239.00	—	—	—	—/—
VideoMAE-B	Dense	1568	356.99	—	—	—	—/—
	20%	1254	313.99	—	—	—	—/—
	40%	941	274.74	—	—	—	—/—
	60%	627	239.00	—	—	—	—/—
TimeSformer-B	Dense	1569	389.81	—	—	—	—/—
	20%	1233	347.15	—	—	—	—/—
	40%	897	305.01	—	—	—	—/—
	60%	561	263.40	—	—	—	—/—

points for VideoMAEv2, VideoMAE, and TimeSformer, respectively. On XD, the corresponding trapezoidal PR-AUC changes are +0.32, +0.61, and +0.68 points. These point estimates support the feasibility of middle-layer reduction with an unchanged head across the tested encoders. They do not imply a universal improvement: all three UCF intervals include both positive and negative changes.

The alternate XD metric matters. Non-interpolated AP changes from 70.33 to 71.30 for VideoMAEv2, 64.86 to 64.51 for VideoMAE, and 70.86 to 72.04 for TimeSformer. In particular, VideoMAE improves under trapezoidal PR-AUC but decreases under AP. We retain both definitions rather than interpret every positive primary-metric change as an unqualified detection gain.

4.3. Budget behavior and reference selections

The 20%, 40%, and 60% deletion settings retain the same insertion depth. For joint attention, the highest setting drops entire pair representatives: a possible explanation for quality changes, not an established breakpoint. TimeSformer’s actual highest deletion is 64.24%; rounding must be reported but is not an isolated causal explanation.

Table 3 places the current method alongside fixed-prefix and fixed-seed random selection within each group. Neither is a global uniform sampler, and random masks are reused across videos. Rankings vary: random is higher on all three UCF configurations, whereas PairSelect is higher on two XD configurations. Thus these references do not establish universal superiority of the norm signal. Their static native-position gathering also differs from the dynamic anchor-slot assignment, so this is a matched-budget reference rather than an isolated norm-only ablation.

4.4. FLOPs and hardware efficiency

Table 4 separates analytical and unmeasured hardware quantities. Principal-block FLOPs decrease by 23.04% (joint attention) and 21.75% (divided attention) at the middle budget, and 33.05% and 32.43% at the highest deletion. Savings are smaller than token deletion because the first six blocks remain dense.

Measurements will fix GPU, input shape, batch size, precision, and attention backend. Full-encoder timing includes embedding, all blocks, selection, gathering, and readout; only diagnostic CPU exports are disabled. The protocol uses 20 warm-up passes, synchronized CUDA events, and three independent process repeats on iden-

tical representative non-test clips. Report throughput at the stated batch, batch-one latency, and peak allocated/reserved memory after synchronized counter resets, together with hardware and software identifiers.

Decoding, host-device transfer, feature writing, and head inference require separate synchronized wall-clock timing. Different clip schedules preclude an end-to-end cross-encoder ranking by clips/s. The dense prefix or selector temporaries may determine peak memory; analytical savings establish neither measured speedup nor memory reduction.

5. SCOPE AND DISCUSSION

Evaluation covers three backbones, two attention families, and one detector family; behavior may vary across architectures, datasets, and metrics. Insertion depth and group size are fixed, not optimized. Standard external reducers such as ATS or ToMe remain to be compared. Earlier diagnostics accessed UCF test data, so development was not wholly test-blind. All additional budgets must be reported, without post-hoc selection implying an independently validated optimum.

Local coverage neither identifies anomalies nor guarantees short-event preservation; anchor positions remain approximate. Reduction affects internal activations, not model weights, output dimension, or detector input count. Deployment benefits depend jointly on measured computation, execution overhead, and hardware behavior.

6. CONCLUSION

PairSelect combines local quotas, magnitude-guided representatives, and non-mixing gathers for joint and divided attention, without changing clip coverage or retraining the pipeline. Completed middle-budget results show small primary-metric changes across three encoders and two datasets, with 21.75–23.04% fewer analytical principal-block operations. The remaining budget and hardware measurements will determine practical quality-efficiency operating points.

7. COMPLIANCE WITH ETHICAL STANDARDS

This work uses the existing UCF-Crime and XD-Violence video benchmarks; it does not report newly collected participant data. **Author confirmation required:** document the applicable dataset permissions and institutional determination of whether ethics approval or exemption is needed. No approval or waiver is asserted in this working draft.

8. ACKNOWLEDGMENTS

Author confirmation required: insert the funding and conflict-of-interest declarations, including an explicit absence-of-conflict statement when applicable. No funding or conflict declaration is inferred here.

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